



## From Internal Combustion to Electric Mobility: A Mixed-Method Investigation of Resilience, Innovation, and Strategic Adaptation among Auto Ancillary Units in the Marathwada Region

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### Keywords:

Electric vehicle transition;  
auto ancillary industry;  
organizational resilience;  
spare parts compression;  
Marathwada region; India  
EV adoption.

### Abstract

The transition from internal combustion engine (ICE) cars to electric mobility is one of the most impactful industrial changes of the 21st century. Although much research has focused on original equipment manufacturers (OEMs) and consumer uptake, the fortunes of the large network of auto ancillary suppliers, especially in developing economies, are poorly understood. This paper examines the challenges that 200 auto ancillary units in the Marathwada region of India are facing to transition from ICE to electric vehicles (EV). Marathwada is a special place as it has a legacy automotive ecosystem of over 1200 ancillary companies, while also offering fantastic opportunities to mega-EV investments by Toyota Kirloskar Motors, Ather Energy and JSW Green Mobility. We then quantify the “spare parts compression effect”—the dramatic reduction in the number of components from hundreds of moving parts in an ICE drivetrain to around 17-20 in an EV—using a survey of 200 companies and semi-structured interviews with 24 companies. According to our data, the 84% of engine/transmission suppliers experienced significant declines in orders (mean 4.6/5 on a 5-point scale). The results from the SEM analysis indicate that the relationship between the resilience enablers (financial reserves, manufacturing flexibility, and workforce skills) and strategic adaptation ( $\beta=0.51$ ,  $p<0.001$ ) is completely mediated via the concept of innovation capability. More than half of micro companies (51%) predict EV products will generate less than 20 percent of sales over the next three years, while more than one in eight (81%) of the larger companies believes they will see sales of more than 20 percent. At this rate, 35 – 40% of the ancillary MSMEs in Marathwada could become obsolete within 10 years if no special measures are taken to address the issue of EV adoption. We are proposing to have a policy roadmap as follows: (i) EV Ancillary Transition Fund with retooling subsidies and skill vouchers; (ii) mandatory OEM supplier development programmes; (iii) “Cluster 2.0” initiative which can make the Marathwada Auto Cluster a hub of capability building. Our results provide practical lessons for industrial clusters that are going through clean technology transitions in emerging economies.

Journal link:  
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## **1. INTRODUCTION**

### **1.1 The Global Automotive Disruption**

For over 100 years, the internal combustion engine has been the primary means of personal and commercial transport. It is a past that is over. There are timelines for phasing out ICE vehicles in place in many countries around the world and there are major automakers joining in on electric futures. The global sales of EVs hit the 17 million mark in 2024, up 35% year-on-year [1]. The momentum is not only in rich countries; emerging markets like India are also undergoing a rapid electrification. But such transitions can not be made easily. They produce winners and losers. The vast majority of the research work on OEM strategies and consumer behaviour, as well as charging infrastructure, has been limited to consumer psychology and behaviour and OEM strategies, while the actual experiences of the thousands of SMEs in the auto components sector has received surprisingly little attention. That is the hidden power behind the auto industry in the form of these niche, family run, slim margins companies for which the specialty are ICE technologies. Whether or not they will adapt will decide whether the EV transition is a social and/or an economic success or failure.

### **1.2 The Auto Ancillary Sector at Risk**

The auto component industry in India expanded to US\$ 80.2 billion with more than 1.5 million direct employees in FY2025 [2]. The estimated MSMEs are 5,000 or more, most of which are micro units. They are closely linked into global supply chains but also massively susceptible to technology change. The EV drivetrain includes an average of 17-20 moving components. Several hundred are found in a modern ICE drivetrain. The number of components being reduced is not just an interesting side note, it's a complete erosion of the existing tooling, engineering knowledge and supplier relationships. As EVs scale up, the proportion of parts required for traditional ICE vehicles that become obsolete is estimated to be between 45 and 84% according to a report by IFEST (2024) [3]. If component suppliers of this type, especially those specialised in ICE, are not able to pivot aggressively, they will face a 50% profit loss by 2030, warns Moody's (2024) [4].

### **1.3 Why Marathwada**

Marathwada is a semi-arid part of eastern Maharashtra, which has been viewed as backward in terms of industry. In the last three decades, however, it's grown to become a highly concentrated

automotive cluster which is based on Chhatrapati Sambhajanagar (Aurangabad) and has other nodes in Jalna, Parbhani, Nanded and Latur. More than 1200 ancillary and engineering units are benefiting from common facilities provided by the Marathwada Auto Cluster (MAC) under the Industrial Infrastructure Upgradation Scheme of Government of India. Meanwhile, Marathwada is fast becoming the most promising EV manufacturing region in India. Toyota Kirloskar Motor has announced a greenfield plant investment of ₹20,000 crore (approx. US\$ 2.4 billion) for a 400,000 electric and hybrid vehicle per annum capacity at Aurangabad Industrial City, AURIC. Ather Energy has installed its third manufacturing plant at AURIC while JSW Green Mobility took over 636 acres for its EV plant [5,6]. The investments are transforming the local industry landscape.

With the legacy ICE supplier base and a rapidly expanding EV OEM ecosystem, Marathwada is a ripe environment for experimentation. What is the reaction of the current ancillary companies? Can they turn, or are they left behind? This paper provides answers to those questions based on primary data of 200 firms.

#### 1.4 Research Objectives

RO1: To assess how the EV disruption affects the auto ancillary units, in terms of perceived impact, within different product segments and firm sizes in Marathwada.

RO2: To determine the factors of resilience that can differentiate the companies that are adapting well from those that are becoming obsolete.

RO3: To analyse strategic adaptation pathways used by these firms such as upskilling, diversification, M&A and open innovation.

RO4: To assess the impact of regional ecosystem factors (MAC, OEM investments, State policies) on enabling/disabling adaptation, and the impact of a rise in India's EV adoption from 6% to 30% by 2036.

#### 1.5 Contributions

This paper has four major contributions. First, it is the first large-scale empirical quantification of the so called 'spare parts compression effect' in an industrial cluster in India. Second, it builds and tests a three-level resilience framework (absorptive, adaptive, transformative) with thresholds in numerical terms. Third, it shows that regional ecosystem moderation has a positive impact on firms that are already resilient, worsening inequality—a 'Matthew effect' that has not

been previously documented in the EV transition literature. Fourth, it provides a policy roadmap for inclusive industrial transition which is differentiated and evidence-based.

## **2. LITERATURE REVIEW**

### **2.1 Discontinuous Technological Transitions**

Not all innovations are created equal. Tushman and Anderson [7] referred to the transition from ICE to EV as a competence destroying technological discontinuity. Skills and tools already acquired are liabilities instead of assets, as are the relationships with suppliers. Christensen's disruptive innovation framework equally holds true: EVs have made an appearance in niche markets (such as e-2Ws and e-rickshaws in India) but are now making their way mainstream. In these transitions, incumbent firms face difficulties as they get stuck in “successful” routines [9]. The auto ancillary sector is just one example. A lot of companies have been around for many years, producing precision engineering pieces. But after all that, battery packs and power electronics aren't much fun.

### **2.2 Organizational Resilience**

Resilience can be defined as the ability to withstand disturbance, reorganize and then carry on functioning [10]. The concept has been further developed and adapted in management studies to incorporate not only recovery but also adaptation and transformation [11,12]. Ortiz-de-Mandojana and Bansal [13] noted that resilient companies make slack resources, have diversified stakeholder relationships and have long-term views. Similarly, a recent bibliometric study by Bhaskar and Kaushik [14] confirmed the increasing trend of interest in organization resilience which stated that majority of the studies were conceptual or qualitative. They called for "large-scale quantitative studies of resilience in specific industrial contexts", which we hope to address in our study. Kumar and Sharma [15] noted that there are three tiers to resilience enablers in the auto ancillary sector: (i) financial and operational buffers; (ii) incremental process improvements; and (iii) radical innovation through collaboration. They did not, however, take into consideration the cluster dynamics in the regions, rather their work was based on a national sample. We extend it through the introduction of a single cluster and the introduction of factors in the ecosystems.

### **2.3 Innovation Capability in the EV Era**

Innovation capability refers to a firm's ability to generate, accept and implement new ideas, processes and products [16]. For ancillary firms, this includes R&D investment, technology

monitoring, external collaboration, and workforce upskilling. The EV transition demands not only new engineering knowledge (high-voltage systems, battery management, power electronics) but also new quality standards such as ISO 26262 (functional safety) and IATF 16949. A study by Wang et al. [17] found that suppliers lacking in-house electronics capabilities faced the greatest difficulties. Another study by Singh and Gupta [18] highlighted the importance of open innovation: firms that collaborated with universities, startups or even competitors adapted faster. Our survey measures innovation capability through nine items (Section B of the questionnaire), allowing us to test its mediating role between resilience enablers and strategic adaptation.

#### **2.4 Spare Parts Compression Effect**

Systematic measurement of the reduction in the number of components between ICE and EV is rare, but it is frequently mentioned in industry reports. The term “spare parts compression” was coined by the International Forum for Environment, Sustainability and Technology (IFEST) in 2024 [3]. They say the number of parts in a typical ICE vehicle is 30,000 to 40,000, with only about 1,200 of those being what they refer to as “active” moving parts in the powertrain. That can be cut down to approximately 150 of the parts are active, and about 17-20 are mechanical moving parts, the rest being electronic. This is a compression that is the lifeline of a piston, valve, camshaft or fuel injector supplier. It's a growth opportunity for a wiring harness or sensor supplier. Our analysis focuses on the differential impact.

#### **2.5 Regional Clusters and Ecosystem Orchestration**

Porter's [19,20] cluster theory states that a group of firms that are linked and clustered together creates competitive advantage through knowledge spillover effects, specialised labour and common infrastructure. Clusters do not necessarily change in response to technology change. They need to be orchestrated, that is, they need to be guided and coordinated by industry associations, governmental bodies or lead companies [21,22]. However, in Marathwada the MAC is the appointed coordinator. It has a common facility centre with tool rooms, testing labs and training. But, as we demonstrate later, the advantages that the MAC offers are not equally enjoyed. Smaller companies may not be aware of the options or may not see the offerings as fitting their needs. This can be understood by the notion of “ecosystem moderation” – the positive effects of well-functioning companies are magnified in the region, but benefits are limited for less well-functioning companies.

## **2.6 India's EV Adoption Trajectory**

The EV market in India has gained momentum from less than 1% in 2018 to 6% in 2025 [23]. Three-wheelers trail the two-wheeler segment and passenger cars come in third. The PM E-DRIVE scheme (replacing the previous FAME II scheme) has provided the central government with ₹10,900 crore for subsidies. The Union Budget 2026-27 added another ₹1,500 crore for the scheme and ₹2,000 crore for charging infrastructure [24]. The demand side incentives and exemption from road tax are provided under the Electric Vehicle Policy 2021, Maharashtra [25]. The state's goal is 10 percent EV registration by 2025, which seems attainable. A more ambitious estimate is offered by NITI Aayog [26] which estimates that EV uptake will increase to 30% in 2036 if the growth is accelerated. The projection is used in our scenario analysis in 4.4.

## **3. THEORETICAL FRAMEWORK**

Three theoretical viewpoints are incorporated.

The resilience capabilities framework identifies three hierarchical levels [27]:

Resilience through absorption – resistance to shock via existing resources (e.g. cash reserves, flexible equipment).

Adjustments to be resilient: incremental changes (e.g. upskill workers, add EV product lines).

Transformative resilience: setting new business models (including buying EV startups, leaving ICE altogether).

Second, from the dynamic capabilities point of view [28,29] there is a focus on sensing, seizing and transforming. The pace of OEM transition plans must be felt by ancillary firms, who must recognise opportunities for investment and/or collaboration and change their organisational identity.

Third, a perspective for orchestration of ecosystems [21,22] considers the regional context. The adaptation of a firm is not an individual process; it is supported by cluster facilities, anchor investments, government policies, and industry associations.

## **4. METHODOLOGY**

### **4.1 Research Design**

We used a sequential explanatory mixed method design [30]. In the quantitative phase, 200 ancillary units were sampled to determine constructs and test hypotheses. Then, the qualitative

phase was used to describe the statistical patterns and surface mechanisms, which included 24 interviews.

## 4.2 Sampling and Data Collection

A sampling frame of 1285 ancillary units, registered with Marathwada Auto Cluster (MAC) and Chamber of Marathwada Industries and Agriculture (CMIA) was used. The sample was stratified by firm size (micro – less than 50 employees, small – 50-199 employees, medium – 200-499 employees, large – 500+ employees) and primary product segment (engine/transmission, electrical/electronics, body/chassis, EV specific, other). We sent 300 questionnaires to firms and received 200 completed questionnaires, a 67% response rate. The data were collected from January to March 2026 through online and paper questionnaires (in English and Marathi).

Table 1. Sample composition (N=200)

| Product segment        | Micro     | Small     | Medium    | Large     | Total      |
|------------------------|-----------|-----------|-----------|-----------|------------|
| Engine/transmission    | 48        | 24        | 8         | 2         | 82         |
| Electrical/electronics | 12        | 14        | 6         | 6         | 38         |
| Body/chassis           | 16        | 12        | 4         | 2         | 34         |
| EV-specific            | 2         | 8         | 10        | 10        | 30         |
| Other                  | 6         | 4         | 4         | 2         | 16         |
| <b>Total</b>           | <b>84</b> | <b>62</b> | <b>32</b> | <b>22</b> | <b>200</b> |

## 4.3 Survey Instrument

The questionnaire was comprised of 33 items in four parts (Appendix A lists the entire questionnaire). The Likert scale items used were on a 5 point scale (1=Strongly Disagree to 5=Strongly Agree).

Section B (15 items): Product segment, firm's EV component sales volume, EV product unit price, sales volume ratio for EV components to other products, direct OEM supply status, EV product price ratio to other products, customer and product awareness, purchase motivation, purchase frequency, satisfaction level with the product, satisfaction level with the supplier, satisfaction level with the company, and need for R&D support.

Section B: Innovation sources (customer, competition, tech trends), enablers (top management support, skilled labour access), barriers (R&D costs, knowledge gaps), and R&D strategy (dedicated budget, external collaboration).

Section C (9 items): EV disruption's effect on order volumes and product mix; Resilience enablers (financial reserves, manufacturing flexibility); Strategic actions (upskilling, M&A, diversification, open innovation).

Section D (8 items): Regional factors: impact of Toyota/Ather/JSW, local leaders (Varroc/Endurance), impact on local EV adoption, state policies, reduction in spare parts, new opportunities with EV parts, impact of MAC adequacy, growth expectations.

Multi-item constructs had an excellent internal consistency, as indicated by Cronbach's alpha ranging from 0.82 to 0.91.

#### **4.4 Qualitative Interviews**

After the survey, we interviewed three groups of 24 semi-structured interviews (45 to 90 minutes each).

16 ancillary firm leaders (representative of high and low resilience and size and segment).

5 ecosystem enablers (CEO of MAC, President of CMIA, banker, official of MIDC, dean of engineering college).

3 OEM and policy stakeholders - Toyota Kirloskar manager, Ather Energy supply chain, Maharashtra Directorate of Industries official)

Interviews were recorded, then transcribed word for word and analysed through thematic analysis [31] in NVivo 14.

#### **4.5 Data Analysis**

The quantitative data was analysed by employing SPSS 28.0 and SmartPLS 4.0. Subgroups were compared using descriptive statistics, ANOVA, and t-tests. All hypothesized relationships were tested using structural equation modelling (PLS-SEM) with 5000 bootstrapped resampling. Model fit was assessed using SRMR (<0.08) and NFI (>0.80).

### **5. RESULTS**

#### **5.1 Descriptive Statistics: The Spare Parts Compression Effect**

Table 2 reports mean scores for key variables by firm size.

The spare parts compression effect (D5) is the highest for micro firms (4.6) and engine/transmission (4.6, not shown in Table 2, but in detailed breakdown). Eighty-four percent of engine/transmission suppliers responded that the decrease of moving parts has had a negative impact on their legacy product lines, with 84% answering “agree” or “strongly agree”.

Barriers to innovation are also highest for micro-firms. The overall average of the technical knowledge of the micros is 4.5 while that of the large firms is 2.9 (L7). The proportion of EV-skilled engineers (B5) is 2.1 for micros and 4.3 for large firms.

There is a stark difference in the resilience enablers. With a micro 1.9 and a large 4.2, these are at a level that is quite secure. These are fairly secure with a micro 1.9 and a large 4.2. Manufacturing flexibility (C4): micro 2.0, large 4.3. C5 – Upskilling investment: micro 1.9, large 4.5.

Table 2 Mean scores (1-5 scale) by firm size

| <b>Variable</b>                    | <b>Micro<br/>(n=84)</b> | <b>Small<br/>(n=62)</b> | <b>Medium<br/>(n=32)</b> | <b>Large<br/>(n=22)</b> | <b>All<br/>(N=200)</b> |
|------------------------------------|-------------------------|-------------------------|--------------------------|-------------------------|------------------------|
| C1: EV reduced ICE orders          | 4.5                     | 4.2                     | 3.9                      | 3.3                     | <b>4.2</b>             |
| C2: Rapid product mix change       | 3.1                     | 3.5                     | 4.0                      | 4.3                     | <b>3.6</b>             |
| B6: R&D/retooling cost barrier     | 4.7                     | 4.5                     | 4.1                      | 3.8                     | <b>4.4</b>             |
| B7: Lack of technical knowledge    | 4.5                     | 4.1                     | 3.4                      | 2.9                     | <b>4.0</b>             |
| B5: Access to EV-skilled engineers | 2.1                     | 2.6                     | 3.6                      | 4.3                     | <b>2.9</b>             |
| C5: Upskilling investment          | 1.9                     | 2.5                     | 3.7                      | 4.5                     | <b>2.8</b>             |
| D5: Spare parts reduction impact   | 4.6                     | 4.3                     | 3.8                      | 3.1                     | <b>4.2</b>             |
| D8: Expect >20% EV growth          | 1.8                     | 2.6                     | 3.8                      | 4.4                     | <b>2.9</b>             |

While 81% of the large firms anticipate revenue increase of more than 20% from EV products in three years, only 12% of the micro firms are expecting the same. (D8>20%). This positive deficit is a warning sign for policy.

## 5.2 Structural Equation Modelling

Measurement model obtained a good fit (SRMR=0.064, NFI=0.85). The convergent and discriminant validity of the results were supported by all factor loadings being above 0.70 and the average variance extracted (AVE) being between 0.53 and 0.68.

Table 3 presents the path coefficients.

Table 3 Structural equation modelling results (N=200)

| Hypothesis | Path                                         | $\beta$ | t-value | p      | Conclusion |
|------------|----------------------------------------------|---------|---------|--------|------------|
| H1         | Resilience enablers → Strategic adaptation   | 0.51    | 8.12    | <0.001 | Supported  |
| H2         | Innovation capability → Strategic adaptation | 0.46    | 7.05    | <0.001 | Supported  |
| H3         | Resilience enablers → Innovation capability  | 0.59    | 9.33    | <0.001 | Supported  |
| H4         | Regional ecosystem → Strategic adaptation    | 0.28    | 4.21    | <0.001 | Supported  |
| H5         | Resilience × Ecosystem → Adaptation          | 0.19    | 3.02    | <0.01  | Moderation |

The strategic adaptation ( $\beta=0.51$ ) is strongly and directly positively affected by the resilience enablers. The indirect effect was mediated to some extent, as  $0.59 \times 0.46 = 0.27$ . Importantly, the moderation effect ( $\beta=0.19$ ) indicates that regional ecosystem factors enhance the positive effect of resilience only for firms that already have some resilience. In other words, the MAC and the OEM investment and government policies favour the more than the less. This “Matthew effect” is among the main findings.

### 5.3 Qualitative Insights

Three major themes emerged from the interviews.

Theme 1: “We are being left behind.” There were a consistent unease and fear from micro engine/transmission owners. One owner (Firm 042) commented, “We know we have to change, but where will the money be coming from? Businesses do not borrow money from banks if they believe the business is dying; our workers have 20 years' experience crankshaft production; they can't design battery trays.” Another (Firm 012) surmised: “The MAC also has great equipment, but what they really need is simple training: how to read the specs of the components they buy for their EVs; how to implement quality systems.”

Theme 2: Divergent adaptation pathways. We were able to see three different routes:

Obsolescence (20-25% of interviewees): mainly small companies, engine and transmission companies, with unclear strategy. They are undergoing passive shrinkage.

Full diversification (50%-60%): large, medium, and small businesses that are selling both EVs and ICE cars, with a target of reaching 60% EV sales by 2030. Full diversification (60%-70%):

all other companies (large, medium, and small) selling both EV and ICE cars, with a 60% EV sales target by 2030. One manager (Firm 089) said: “No, we're not going to become EV-only businesses overnight. We're going to be 60% ICE, 30% EV, 10% aftermarket. The problem is that EV customers have different quality standards.”

Radical transformation (25-30%): primarily large companies and electrical and electronics specialists. For instance, Varroc started its EV lighting and battery management system (BMS) manufacturing in 2022 and is now working in ADAS components.

Theme 3: Ecosystem ambivalence. The MAC's common facility centre was commended by large companies, particularly the testing laboratories. A quality manager of the firm (Firm 056) stated: “Without the MAC lab we would be sending samples to Pune taking weeks, the lab is a must.” However, micro firms were less likely to be aware of training programmes or that programmes were too far advanced. The outreach to the smallest firms seems to fall short of the MAC.

#### **5.4 Projecting 6% to 30% EV Adoption**

Representatives of the OEMs admitted the strain. Toyota Kirloskar manager said, “We are working on getting the local supply chain developed, but with EVs, there are more precision and safety requirements that some existing suppliers can meet while the others can't.

Based on the accelerated scenario from NITI Aayog [26] India will achieve 30% EV penetration by 2036. We estimate, based on our survey data:

Demand for ICE-specific components will decline by 55-70% from 2025 levels.

A combined revenue loss of something like ₹1200-1800 crore (approx. US\$ 140-210 million) for a sample of 82 engine/transmission manufacturers (41% of total) is the total loss if they don't diversify.

37% of companies have low adaptation readiness (composite score  $\leq 9$  out of a range of 9 to 27). This group if not improved, may exit by 2035, with 70-80 micro firms (35-40% of sample) exiting.

On the other hand, companies in the field of electrical/electronic and EV could expand by 25-40% a year.

These forecasts are not the only fearsome ones, but they are terrifying. They highlight the need for specific policy intervention.

## 6. DISCUSSION

### 6.1 Theoretical Contributions

Contribution 1: Defined the three-tiers of resilience quantified. We set numeric values for absorptive, adaptive and transformative resilience. Only 24% of micro firms have absorptive resilience ( $C3+C4 \geq 3.5$ ) while 91% of large firms do. Adaptive resilience ( $C5+C7 \geq 3.5$ ) 11% micro vs. 86% large. Transformative resilience ( $C6+C9 \geq 3.5$ ) 3% micro versus 68% large. These numbers represent not just a description, but can inform policy targeting.

Contribution 2: Experimental verification of the compression of spare parts. As far as we know this is the first study that has measured the spare parts compression effect within the firm level in India. The 4.2 average score on D5—4.6 on engine/transmission companies—is an indication of industry concerns and a measurement of the existential threat.

Contribution 3: Inequality is enhanced by the moderation of the ecosystem. The significant interaction term ( $\beta=0.19$ ) indicates that regional ecosystem resources (MAC, OEMs, policies) impact firms more when they are already in a resilient state. This is a cautionary discovery: cluster level support has the potential to exacerbate inequities if intentionally not planned for.

### 6.2 Policy Implications

From our evidence, we make three specific proposals for policies.

The EV Ancillary Transition Fund (EATF). A ₹500 crore corpus for 5-year period, managed by MIDC and MAC including:

Subsidies for micro/small engine/transmission companies for 40-60% of capital expenditures.

Confirmed orders for EV components will receive working capital loans at 4% interest.

Skill voucher of Rs 50,000 per worker for EV Electrical and Software skill training.

2. OEM-mandated Supplier Development. The anchor OEMs (Toyota, Ather, JSW) are required to:

Procurement spend 2% or more of the annual procurement spend with micro/small local suppliers.

Offer technology transfer and support for ISO 26262 certification.

Provide space in the 'second-tier supplier park' at AURIC for MSME consortia.

3. MAC 2.0 – Capability-Building Hub. Make the MAC common facility centre a 'capability centre', rather than a common facility centre:

Introduce a mobile training unit which visits 500 micro firms every year.

Certification Cell for IATF 16949 and functional safety.

Establish a common prototype lab for EV components (battery trays, busbars, thermal systems) with subsidised access for micro-firms.

These are recommendations that are actionable and based on our quantitative evidence.

### **6.3 Managerial Implications**

Our results provide a clear guidance for managers of ancillary firms:

Invest in Skills prior to machines. The individual effect of upskilling (C5) on strategic adaptation was the strongest ( $\beta=0.31$ ,  $p<0.001$ ), not presented in Table 3 but in detailed output. For every ₹ invested in trainings, the capacity to adapt is about ₹3-5.

Pursue open innovation. Those that worked with engineering colleges, start-ups or even with competitors (B9) adapted very well.

Toss the coin and don't wait for the answer! Investment in early, albeit imperfect, EV capabilities by firms has yielded positive results, whereas later investments have had negative results.

### **6.4 Limitations and Future Research**

There are some limitations to this study. First, the cross-sectional design does not lend itself to making strong causal statements. It would be useful to have a longitudinal panel. Second, we used self-reported perceptions which would be strengthened by objective data (e.g., financial statements, patent counts). Third, the study is limited to Marathwada only; further studies are needed to generalise the results across other Indian clusters (Pune, Chennai, NCR). Fourth, our sample may have been biased towards the larger micro firms as they are more easily contacted. More research is needed focusing on a cohort of firms over time, adding financial performance measures, and using comparative studies between different clusters. Additionally, the effect of gender and social capital on resilience needs to be explored.

## **7. CONCLUSION**

The shift from IC engine to electric mobility is a necessary and inevitable change in the interest of environment. Yet, its social and economic impact on auto ancillary workers and owners must not be overlooked. Our analysis of 200 companies in Marathwada shows a split picture—some large companies already ready for EVs are set to grow while many micro and small companies are specialised in ICE are at risk of extinction.

The adoption rate of EVs in India is expected to go up from 6% to 30% by 2036. We believe that in the next 10 years, if the trend continues as it is, 35 to 40 % of the ancillary MSMEs in

Marathwada will shut down leading to significant job losses. But this is not a given. An inclusive transition is possible with a dedicated transition fund, OEM-led supplier development and a re-imagined cluster capability centre. In the next 2 to 3 years, the decisions taken by the policymakers and industry leaders will decide if Marathwada is a case of a mere industrial transformation or a cautionary tale. We believe the latter is possible from our evidence.

## **ACKNOWLEDGMENTS**

The author is grateful to Marathwada Auto Cluster, the Chamber of Marathwada Industries and Agriculture (CAIA) as well as all the participating companies and individuals. All the faculties of Department of management Science from Dr. Babasaheb Ambedkar Marathwada University is acknowledged.

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